

Bullet Separator Examples

Example 1 : Set Comprehension with Bullet

The bullet separator distinguishes the constraint (filter) from the term (selector):

$$\{x : \mathbb{N} \mid x > 0 \bullet x * x\}$$

This reads: "the set of x squared, for all x in $\mathbb{N}$ where x is positive". The constraint $x > 0$ filters the domain, and $x * x$ is the value selected.

Example 2 : Set Comprehension Without Bullet

Without the bullet, the set contains the values that satisfy the predicate:

$$\{x : \mathbb{N} \mid x > 0 \wedge x < 10\}$$

This is the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. No transformation is applied.

Example 3 : Comparing With and Without Bullet

With bullet (maps/transforms values):

$$\{n : \mathbb{N} \mid n < 5 \bullet n * n\}$$

Result: $\{0, 1, 4, 9, 16\}$ (squares of 0, 1, 2, 3, 4)

Without bullet (filters values):

$$\{n : \mathbb{N} \mid n < 5\}$$

Result: $\{0, 1, 2, 3, 4\}$ (the values themselves)

Example 4 : Complex Set Comprehension

$$\{x : \mathbb{Z} \mid x \bmod 2 = 0 \wedge x \geq -10 \wedge x \leq 10 \bullet x \div 2\}$$

This gives the set $\{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$ by taking even integers from -10 to 10 and dividing by 2.

Example 5 : Mu Operator with Bullet

The mu operator finds the unique value satisfying a condition:

$$(\mu x : \mathbb{N} \mid x > 5 \wedge x < 7 \bullet x)$$

This evaluates to 6, the unique natural number strictly between 5 and 7. The bullet separates the uniqueness constraint from the selected expression.

Example 6 : Mu with Complex Expression

$$(\mu n : \mathbb{N} \mid n * n = 16 \bullet n + 1)$$

This finds the unique n where n squared equals 16 (which is $n = 4$), then evaluates $n + 1$, giving 5.

Example 7 : Nested Expressions

$$\{x : \mathbb{N} \mid x < 3 \bullet \{y : \mathbb{N} \mid y < x \bullet (x, y)\}\}$$

This creates a set of sets of tuples. For each $x < 3$, we create the set of pairs (x, y) where $y < x$.

Example 8 : Quantifiers with Bullet (Limited Support)

Standard Z notation supports bullet in quantifiers to separate constraints from body:

PROPOSED: $\forall x : \mathbb{N} \mid x > 0 \bullet x \geq 0$

This would read: "for all positive x, x is non-negative". The constraint $x > 0$ restricts the quantification domain.

NOTE: Full bullet support in forall/exists quantifiers is not yet implemented. Currently use implication instead:

$\forall x : \mathbb{N} \bullet x > 0 \Rightarrow x \geq 0$

Example 9 : Partial Functions with Bullet

In pattern matching and function definitions:

$$\frac{\text{predecessor} : \mathbb{N} \rightarrow \mathbb{N}}{\forall n : \mathbb{N} \mid n > 0 \bullet \text{predecessor}(n) = n - 1}$$

The bullet indicates predecessor is only defined when the constraint $n > 0$ holds.

Example 10 : Practical Example - Data Filtering

[Employee]

$$\frac{\begin{array}{l} \text{salaries} : \text{Employee} \rightarrow \mathbb{N} \\ \text{high_earners} : \mathbb{P} \text{Employee} \\ \text{high_salaries} : \mathbb{P} \mathbb{N} \end{array}}{\begin{array}{l} \text{high_earners} = \{ e : \text{Employee} \mid \text{salaries}(e) > 100000 \} \\ \text{high_salaries} = \{ e : \text{Employee} \mid \text{salaries}(e) > 100000 \bullet \text{salaries}(e) \} \end{array}}$$

high_earners is the set of employees earning over 100K (the employees themselves). high_salaries is the set of salary values over 100K (the transformed values).

Example 11 : Mathematical Sets

The set of perfect squares less than 100:

$\{ n : \mathbb{N} \mid n < 10 \bullet n * n \}$

Result: $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$

The set of even numbers:

$\{ n : \mathbb{N} \mid n \bmod 2 = 0 \}$

Result: $0, 2, 4, 6, 8, \dots$ (infinite set, no bullet needed)

Example 12 : Design Pattern

The bullet separator follows a consistent pattern across Z notation:

Before bullet: constraint (declaration and filter) After bullet: term (what to compute or select)

This separation makes specifications clearer by explicitly distinguishing the domain restriction from the value computation.